

CHRIST
(DEEMED TO BE UNIVERSITY)
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Hobbit
(Hobby and Activity-centric Social Media)

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1.0 Introduction

1.1 Purpose

Hobbit is an activity-based social networking platform designed for hobbyists and enthusiasts. Existing social media platforms encourage users to showcase achievements without providing meaningful ways to document their interests and personal growth. Hobbit aims to solve this problem by allowing users to track and share their hobbies and activities.

The platform enables users to log:

- Books they read
- Movies and TV shows they watch
- Games they play
- Music they listen to
- Sports they participate in
- Walks and fitness activities
- Other personal interests and hobbies.

The platform combines the concepts of Letterboxd,

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Goodreads, Last.fm, Strava and Serialzed into one unified ecosystem.

1.2 Scope

The scope of the project includes:

- User registration and authentication
- Activity logging system
- Personalized activity feeds
- User profiles
- Activity statistics and insights
- Following and social interactions
- Real-time messaging
- Activity recommendation system
- Cross-domain activity tracking

The application targets:

- Readers
- Gamers
- Movie enthusiasts
- Music lovers
- Fitness enthusiasts
- Sports participants
- Hobby communities.

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1.3 Stack Descriptions

The application is designed for extremely high scalability and performance.

Frontend
- Typescript
- React
- Next.js
- Tailwind CSS

Backend
- Rust
- Axum Web Framework
- Tokio Async Runtime

API
- gRPC
- REST APIs
- Websockets

Databases
- PostgreSQL
- Redis
- Clickhouse

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- Elasticsearch
- Object Storage (S3)

Infrastructure

- Docker
- Kubernetes
- NGINX
- Cloudflare CDN
- Prometheus
- Grafana

Development Tools

- Visual Studio Code
- Git
- GitHub
- Postman

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2.0 OVERALL DESCRIPTION

2.1 Requirement

The Hobbit platform is required to provide a centralized digital ecosystem where hobbyists can document, share, and discover activities.

The application should:

- Support millions of users.
- Provide low-latency feeds.
- Handle real-time messaging.
- Generate personalized recommendations
- Store activity history indefinitely.
- Support multiple activity types.
- Offer analytics and statistics.

The system should remain highly available and fault tolerant.

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2.2 Module Description

1) Feed
2) Profile Page
3) Activity Overview Page
4) Activity Statistics
5) Activity Type Display Page
6) Activity Blend Page
7) Following Feed
8) Chats

↳ Blends together all necessary social media features with activity statistics.

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3.0 Functional Requirements

3.1 Software Requirements

Operating System: Linux Server

Browser Support: 1) Chrome
2) Firefox
3) Edge
4) Safari

Development Environment:
1) Node.JS 2) Rust Toolchain
3) Docker engine 4) Kubernetes

Dependencies:
1) PostgreSQL 2) Redis
3) Elasticsearch 4) Clickhouse

3.2 Hardware Requirements

Development Machine
1) Processor: Intel i5 or Apple M1
2) RAM: 8 GB minimum
3) Storage: 50 GB SSD

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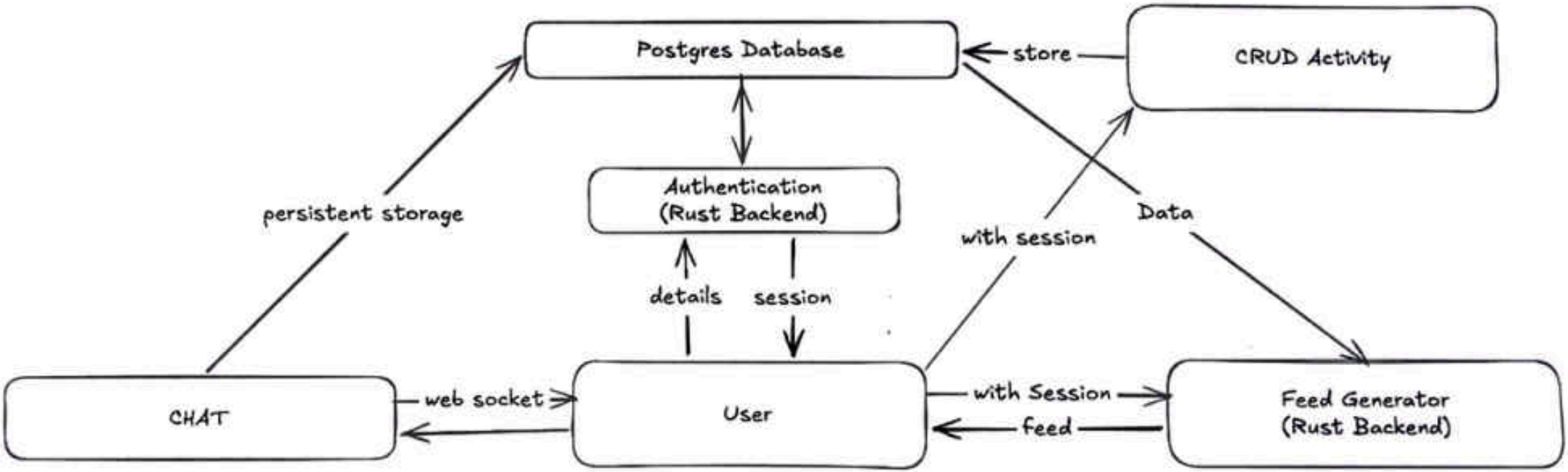
Production Environment:

1) Processor: 32+ CPU Cores
2) RAM: 128 GB
3) SSD Storage: 2TB

4.0

Table Description

Module	Database	Description
Users	PostgreSQL	Stores user data
Activities	PostgreSQL	Stores posts/activities
Feed	Redis	Caching the feed
Search	Elasticsearch	All-site search index
Analytics	ClickHouse	Activity time-series analytics
Media	Object Storage	Images
Notifications	~~Redis~~ Kafka	Notification streaming


Postgres Database
store
CRUD Activity
Authentication
(Rust Backend)
persistent storage
Data
with session
details
session
CHAT
web socket
User
with Session
feed
Feed Generator
(Rust Backend)

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5.0 Data flow Diagram

Please refer to the
the otherside

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6.0 Web Application Deployment

1) Containerize Application
2) Deploy containers to Kubernetes
3) Deploy Database containers
4) Configure NGINX reverse proxy
5) Set up CDN
6) Set up monitoring, CI & CD.

7.0 References

1) Rust-lang.org
2) nextjs.org
3) postgresql.org
4) clickhouse.com
5) redis.io
6) grpc.io
7) prometheus.io
8) grafana.com
9) docker.com
10) Kubernetes.io.